

AMAZON SDE 2 MONTH INTERNSHIP EXPERIENCE 2022

About the company:

Amazon is a product based company. It has a lot of divisions like e-commerce, cloud computing (AWS), digital streaming, AI (Alexa), ECHO and so on. It's one of the MAANG companies.

Experience:

Amazon isn't a regular visit to our college. They approached our college for internship 2 years back and left without picking any interns. But in recent trends, Amazon is taking a good count of interns from all the colleges around India.

They held two rounds for selection. The first one was online assessment. They asked 2 coding questions and then a work personality test. The questions were very easy, it was more of practical and implementation oriented. Most of us were able to attend both of them. The personality test was kept to check your work ethics and whether you are a fit for their work culture. It is almost like other usual personality test where you would be provided with a situation and you must either agree or disagree. We still don't know whether it is taken as a metrics for evaluation.

QA: (HackerRank)

The question pattern for the online assessment would change for every year. This year it was only coding round whereas previously they would ask coding questions along with code debugging (i.e semantic and syntax errors) and some technical questions sections. So prepare yourself for both coding and also technical questions.

Interview:

Then for the interview, they selected around 74 students (both MIT and CEG). The interview is one of the crucial part in selection for Amazon.

Interview Format: Code pair

"I'd rather interview 50 people and not hire anyone than hire the wrong person" - Jeff Bezos.

In the interview, we had an introduction where each of us took a turn introducing ourselves. In this interview, you wouldn't have any HR round to talk about your strengths. So when you introduce yourself, tell them your technical strengths and what projects you have worked on. Present yourself in

a pleasing way. Then you would be asked to share your screen and use **LiveCode** to code your questions.

You will be provided with a question. Even if you already know the answer, start with brute force because the entire interview will be testing your communication with the interviewer and your approach to problems.

When providing your approach, go in hand with the time and space complexity and also why you are going with it. And then, after the approach, you will be asked to code. When doing so, explain the code when you are typing and what that snippet would work as.

If time permits, the interview would go for another question. In some cases, 3 questions could be asked. It depends on the interviewer. In the end, you will be allowed to ask a few questions. Ask appropriate questions that help them understand how much you are interested in working for the company.

Round 1 - Online Assessment

The Online Assessment was split into two sections. One being the coding round and other being the work-personality round.

The coding round was really easy. The question were like real life problems of the basic question.

1. Optimize arranging the letters in a button cell phone.



For each string given as test case, we have to find the optimized way to arrange the letters in the 9 button such that minimum number of clicks should be performed to type the text.

Sol: use Priority Queue

2. Find the grayest area in a given screen (matrix) .

Each cell in a matrix/ screen may be black which is represented as -1 and white as 1. To calculate grayness for each cell you would have to find the

difference between no of black area and no of white areas in the cell's identified row and col. The one with maximum value (absolute) is the answer.

Sol: Calculate row_sum and col_sum for each row and col. Then using that, find the maximum absolute sum of row_sum[i] and col_sum[i].

Round 2 - Code pair interview

1. Run-length-encoding

<https://www.geeksforgeeks.org/run-length-encoding/>

The task is to convert a string "aaabbcccc" to "a3b2c4"

Questions asked

- Condition in which a string could be "aaaabbbaa"
- Reduce space complexity
- Does the length of the solution string is always less than original

2. Happy Strings

<https://leetcode.com/problems/longest-happy-string/>

The task is you will be given 3 characters 'a', 'b', 'c' that has a, b, c occurrences. The substring of the resultant string generated shouldn't have three consecutive same characters (like "aaa"). Generate the longest string

Question asked

- The same question when instead of three characters infinite number of numbers would be provided with any number of occurrences.

Tips:

Prepare well on these topics

- Strings
- Linked List
- Trees
- Arrays

The online assessment needs practice on coding. Prepare using LeetCode, GeeksForGeeks. The interview shortlist process might need a bit of luck. But when you enter the interview, you should be able to tackle the trick questions posed by the interviewer. In the end, after all the preparation go through the 14 leadership requirements from Amazon (just in case, interviewer may check) and STAR approach.